

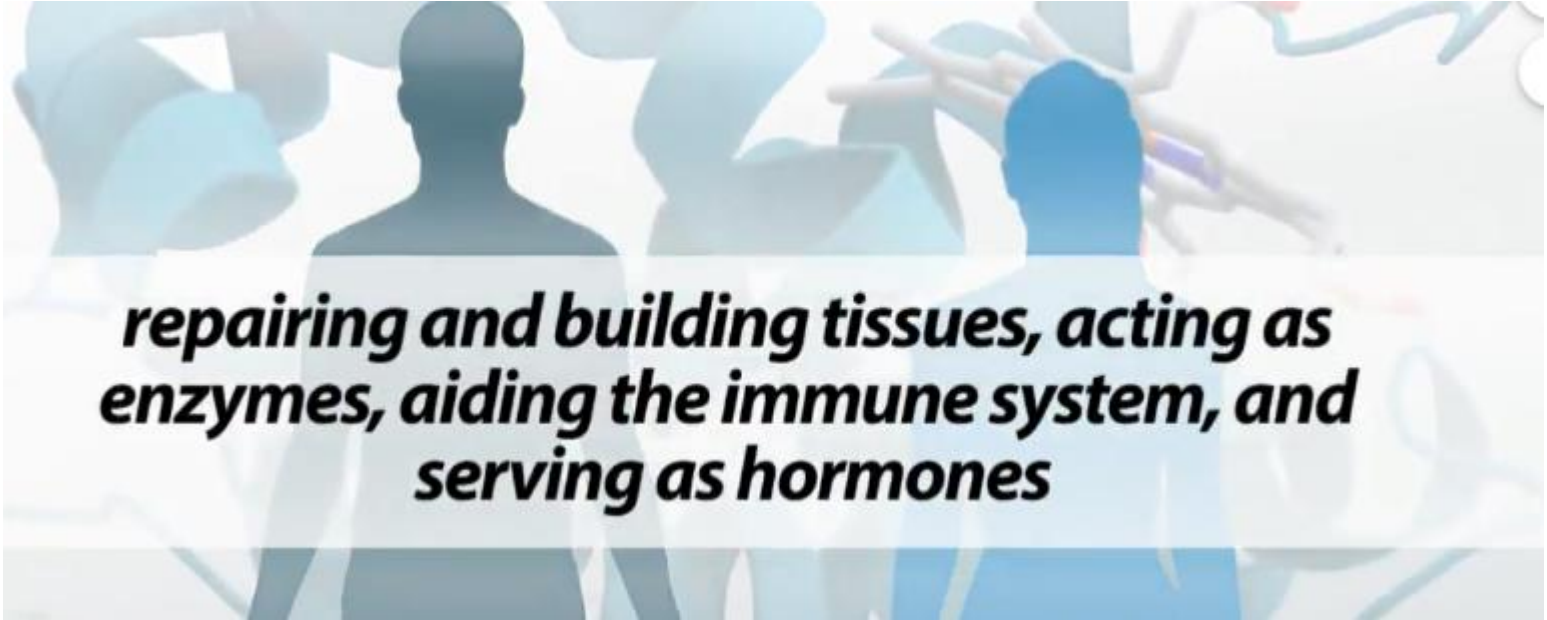
Protein chemistry

Prof. Dr. Mohamed Lebda

Protein

- Proteins are large, complex organic molecules that are critical for the normal functioning of the body.

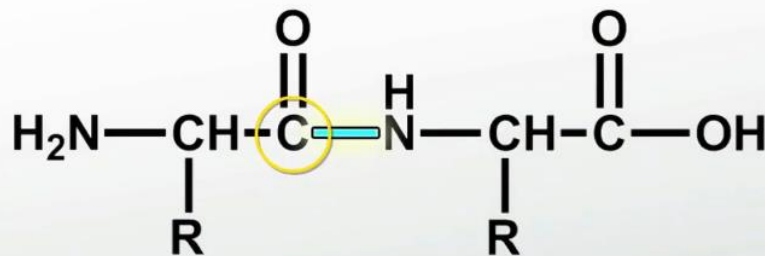
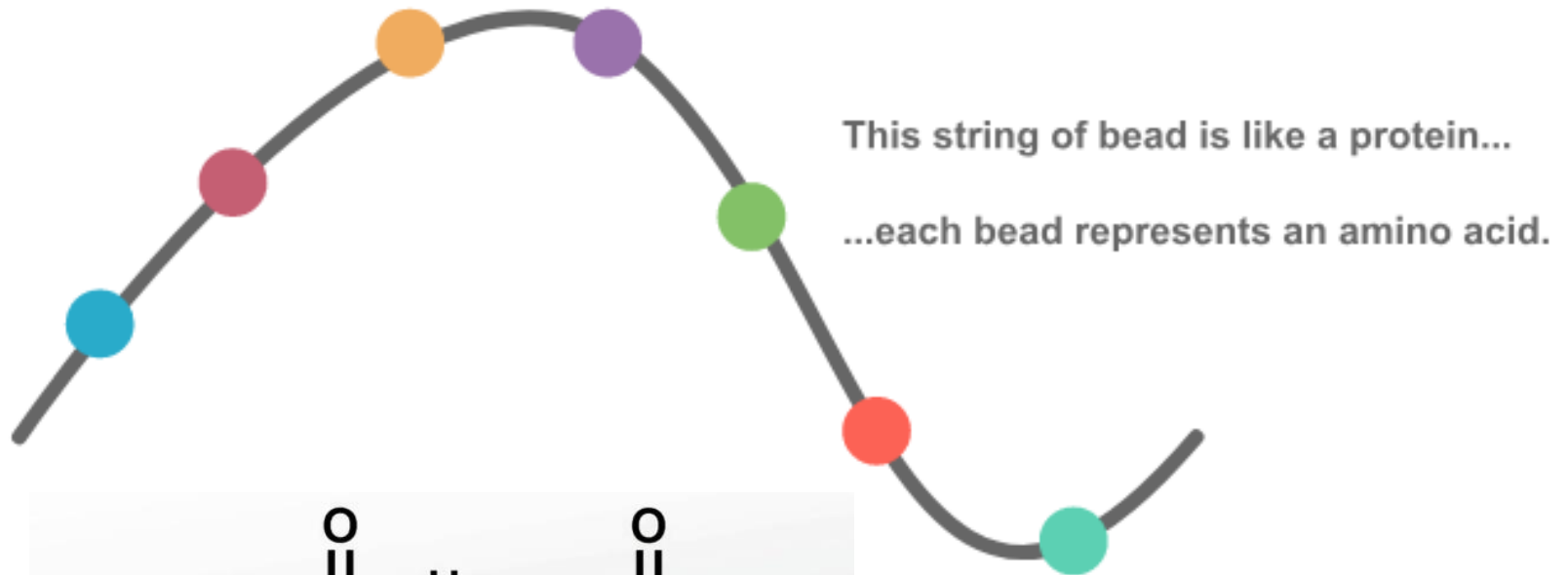
General function of protein

An illustration featuring the silhouettes of several people in various poses, overlaid with a complex, colorful molecular structure. The background is a light, neutral tone.

repairing and building tissues, acting as enzymes, aiding the immune system, and serving as hormones

Protein

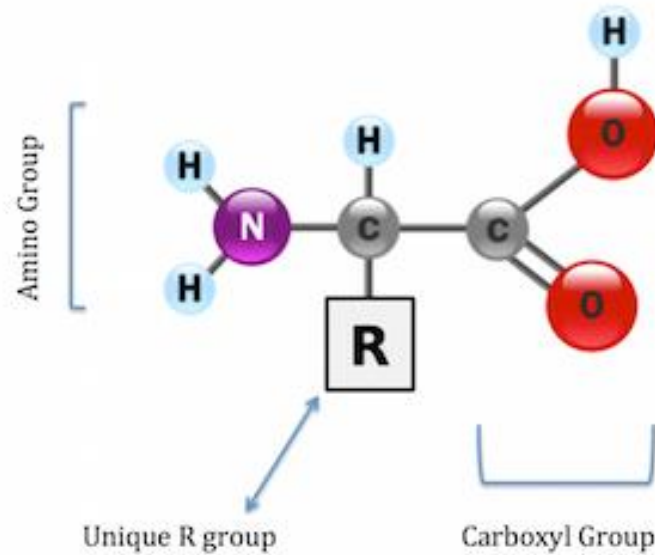
- Proteins are made up of hundreds of smaller units called amino acids that are attached to one another by peptide bonds, forming a long chain



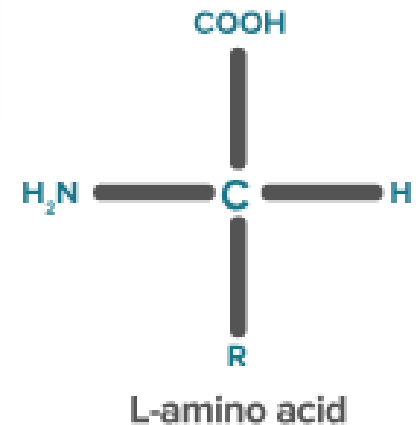
Peptide Bond
the covalent bond that
holds amino acids together

Amino acids

- General structure



The Structure of an Amino Acid



- All amino acids at a pH 7 are L-amino acids
- All amino acids at a pH 7 are amphoteric
- All amino acids (except glycine) are optically active
- Although about 300 amino acids found in nature, only 20 enter in protein structure (primary).

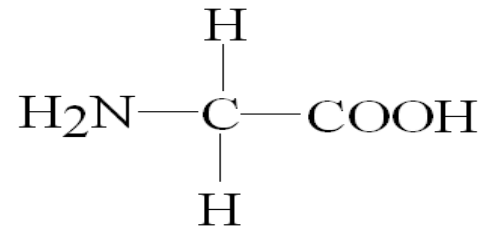
Classification of amino acids

According to chemical structure

I- Aliphatic amino acids: having no ring structure

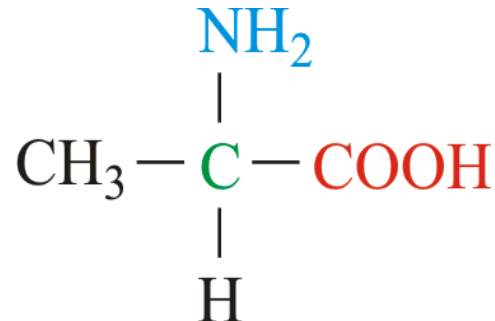
A- *Neutral amino acids* : contain one —COOH group and one —NH_2 group

Glycine: (Gly = G)



glycine

Alanine: (Ala = A)



Classification of amino acids

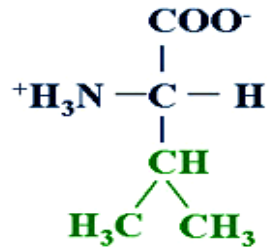
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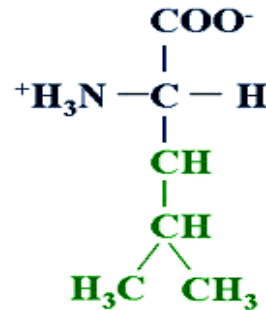
A- *Neutral amino acids* : contain one —COOH group and one —NH_2 group

Branched chain amino acids:

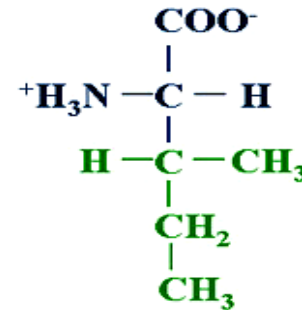
1. **Valine:** (Val = V)
2. **Leucine:** (Leu = L)
3. **Isoleucine:** (Ile)



Valine



Leucine



Isoleucine

Classification of amino acids

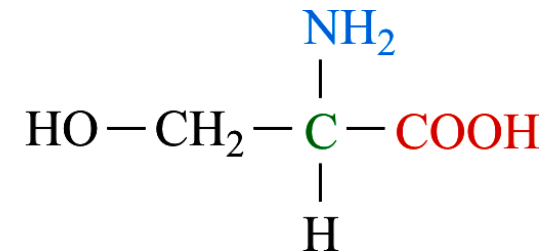
According to chemical structure

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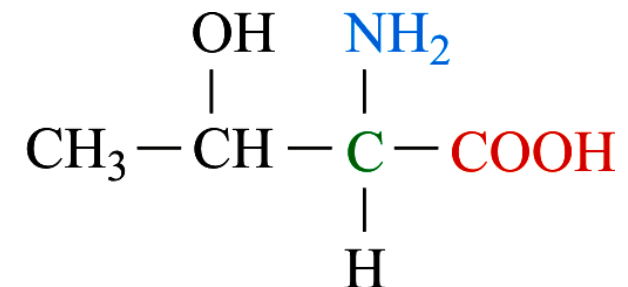
A- *Neutral amino acids* : contain one —COOH group and one —NH_2 group

Hydroxy containing amino acids:

1. **Serine:** (Ser = S)



2. **Threonine:** (Thr = T)



Classification of amino acids

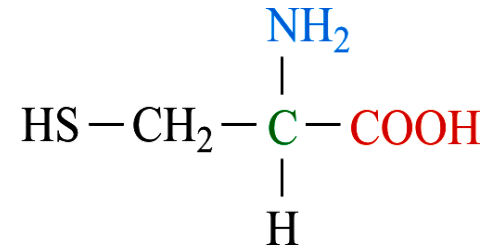
According to chemical structure

I- Aliphatic amino acids: having no ring structure

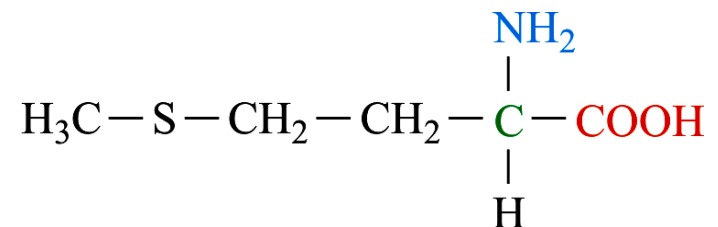
A- *Neutral amino acids* : contain one —COOH group and one —NH_2 group

Sulfur containing amino acids:

1- Cysteine (Cys = C)



2- Methionine: (Met = M)



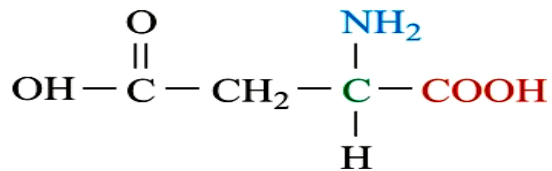
Classification of amino acids

According to chemical structure

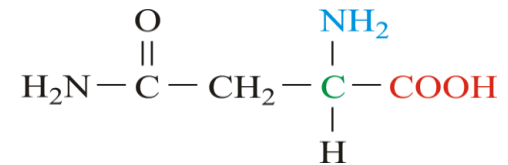
I- Aliphatic amino acids: having no ring structure

B- acidic amino acids : contain one amino group and more than one carboxylic group

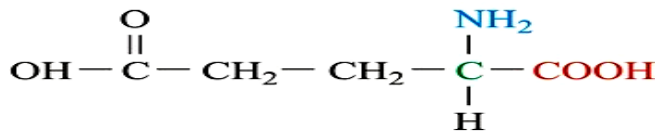
1- Aspartic acid: (Asp = D)



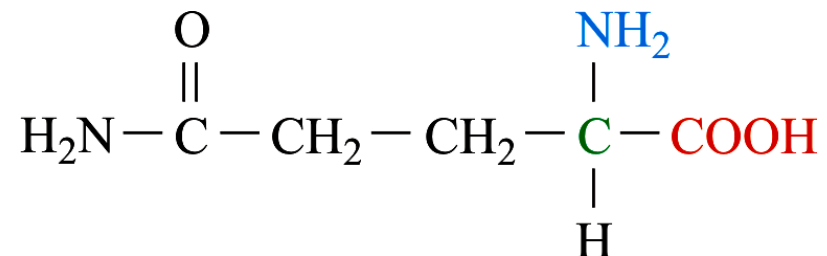
2- Asparagine



3- Glutamic acid: (Glu = E)



4- Glutamine



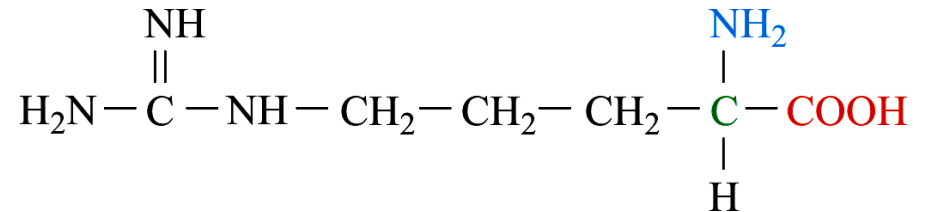
Classification of amino acids

According to chemical structure

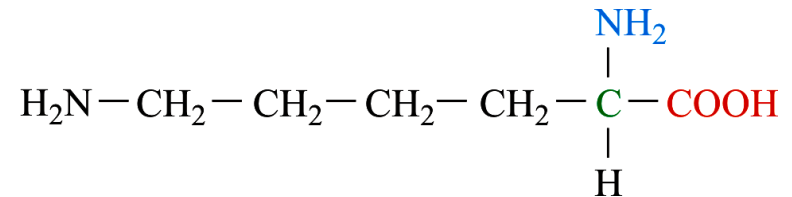
I- Aliphatic amino acids: having no ring structure

C- Basic amino acids : contain one carboxylic group and more than one amino group

1- Arginine: (Arg = R)



2- Lysine: (Lys = K)

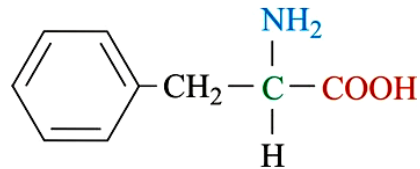


Classification of amino acids

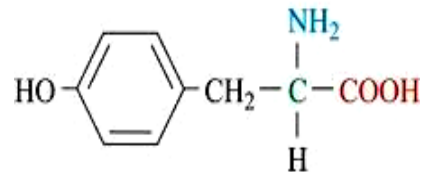
According to chemical structure

II- Aromatic amino acids : contain aromatic ring

1- Phenylalanine: (Phe = F)



2- Tyrosine: (Tyr= Y)

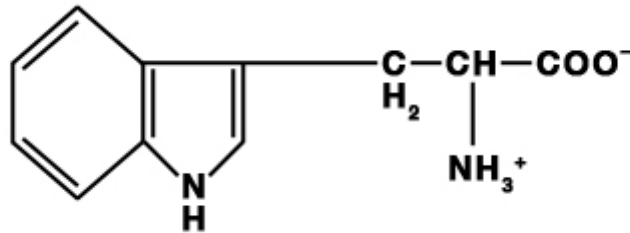


Classification of amino acids

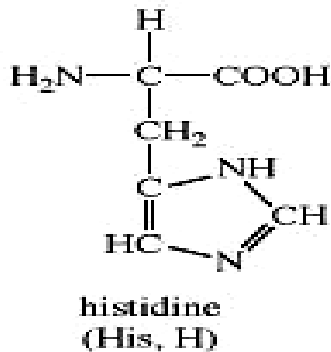
According to chemical structure

II- Aromatic amino acids : contain aromatic ring

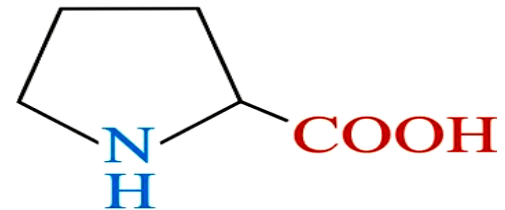
3- Tryptophan: (Trp = W)



4- Histidine: (His = H)



5- Proline : (Pro = P)



Classification of amino acids

According to the optical activity

- A. D-amino acids: with the amino group on the right of asymmetric carbon atom.
- B. L-amino acids: with the amino group on the left of asymmetric carbon atom.
- C. None optically active amino acid.

Classification of amino acids

According to the nutritional value (biological value)

Essential (indispensable) amino acids	Non essential (dispensible) amino acids
Can not be synthesized by body	Can be synthesized in the body
Their deficiency in diet affect growth and health	Their deficiency in the diet doesn't affect growth and health
They are valine, leucine, isoleucine, phenylalanine, tryptophan, methionine, lysine, arginine, histidine and threonine	They include glycine, alanine, serine, cysteine, asparagine, aspartic acid, glutamic acid, glutamine, tyrosine and proline

N.B. Arginine and histidine are called semi-essential

Classification of amino acids

According to the metabolic fate

1- Ketogenic amino acids:

- Give rise to ketone bodies and fat during its catabolic pathways
- e.g. leucine.

2- Glucogenic amino acids:

- Give rise to glucose during its catabolic pathways
- e.g. glycine, glutamine, proline, serine, alanine, threonine, aspartic acid, asparagine, glutamic acid, methionine, cysteine, arginine, valine and histidine.

3- Ketogenic and glucogenic (mixed) amino acids:

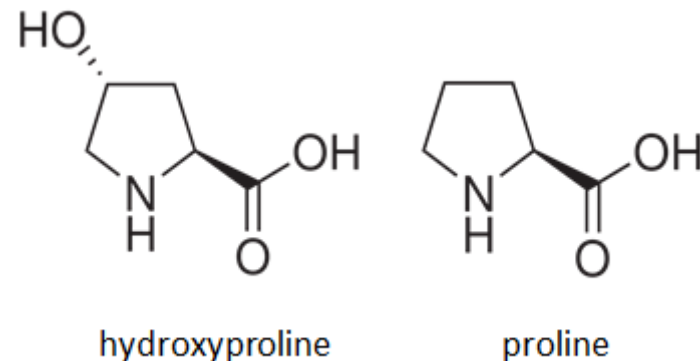
- Give rise to either fat or glucose during its catabolic pathways
- e.g. isoleucine, lysine, tryptophan, phenylalanine and tyrosine.

Derived or modified amino acids

- Non- primary amino acids found in proteins
- After the synthesis of proteins, some of the amino acids are modified in post-translation processing.

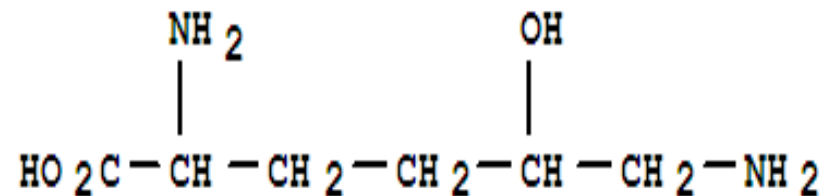
1. 4-hydroxyproline

- found in collagen
- formed by hydroxylation of proline.



2. 5-Hydroxylysine

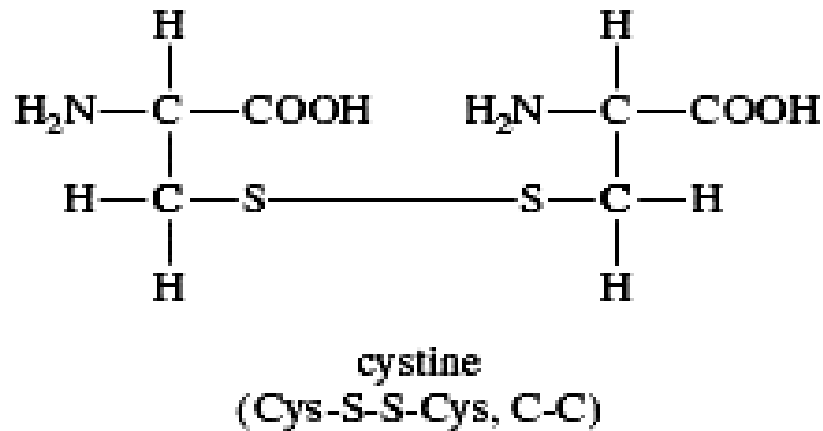
- found in collagen
- formed by hydroxylation of lysine.



Derived or modified amino acids

3. Cystine

- formed by conjugation of two cysteines via disulfide linkage
- found in high concentrations in digestive enzymes

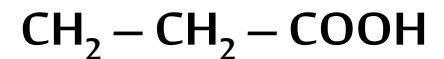


Non protein and Non α -amino acids:

- These are amino acids that do not occur in proteins
- Perform other functions in metabolism

1- β -alanine:

- a component of COA, carnosine
- Catabolic product of some pyrimidine bases.



2- γ -aminobutyric acid (GABA)

- formed from glutamic acid
- Neurotransmitter



Non protein and Non α -amino acids:

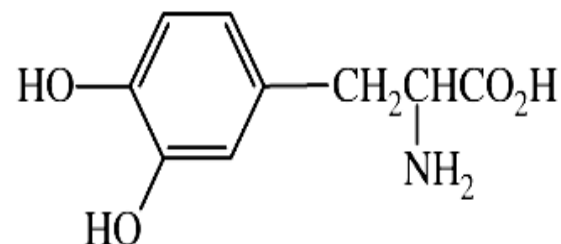
3- Taurine:

- occurs in bile combined with bile acids



4- DOPA (3,4-dihydroxyphenylalanine)

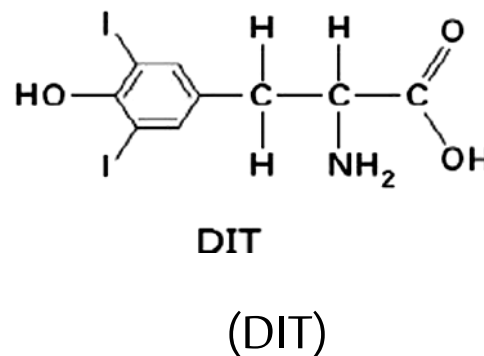
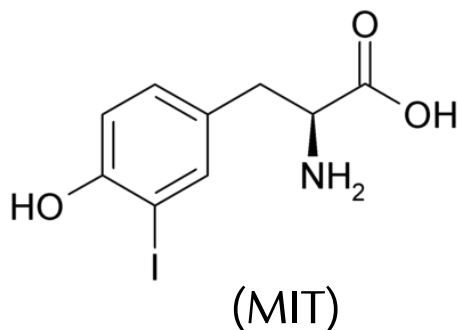
- precursor for melanin pigment, epinephrine and norepinephrine



L-DOPA

5- Monoiodotyrosine (MIT) and Diiodotyrosine (DIT)

- Precursors of thyroid hormones (T3 and T4)



Amino acids

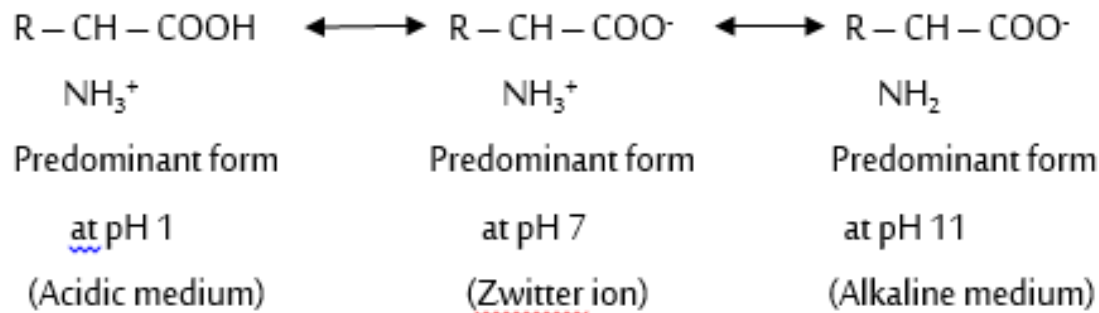
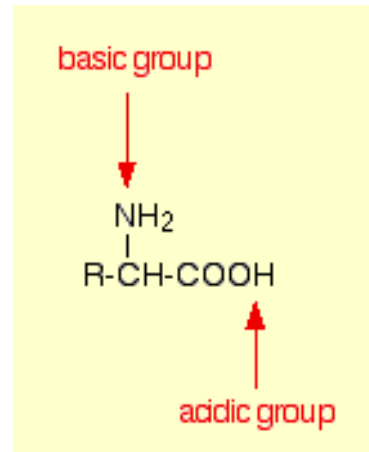
- **General function**

1. Building units of proteins e.g. tissue protein, plasma proteins, all enzymes and protein hormones.
2. Building units of peptides e.g. glutathione
3. Act as neurotransmitters i.e. glycine and glutamate .
4. Used in detoxification reactions e.g. glycine.
5. Precursor for synthesis of important biological compounds e.g. histamine, γ -aminobutyric acid (GABA) , creatine, T3, thyroxine, epinephrine, melatonin, serotonin, S-adenosyl methionine.

Properties of amino acids

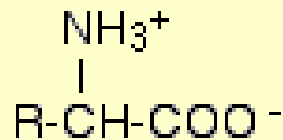
1. Physical properties of amino acids

1. **Solubility:** All amino acids are soluble in **water, diluted acids and alkalis**
2. **Optical activity:** All amino acids **except glycine** are optically active
3. **Amphoteric property:** All amino acids behave as acids and alkalis (Zwitter ion)



Properties of amino acids

Zwitter ion :is a compound with **no** overall electrical charge, but which contains separate parts which are positively and negatively charged. e.g. amino acids



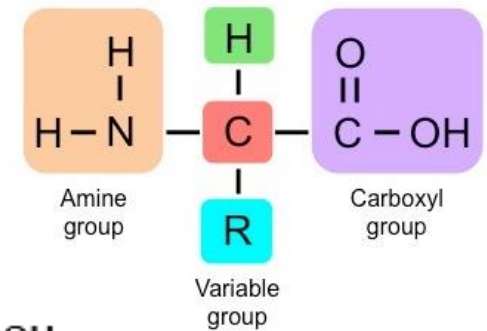
a zwitterion

Isoelectric point (IEP or IP) : The pH at which an amino acid carries **equal** positive and negative charge e.g. IP of glycine is pH 6.07

Properties of amino acids

II. Chemical properties of amino acids:

A. Reactions due to presence of **amino group** of amino acids:

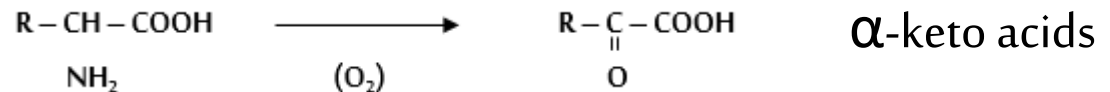


1- Salt formation

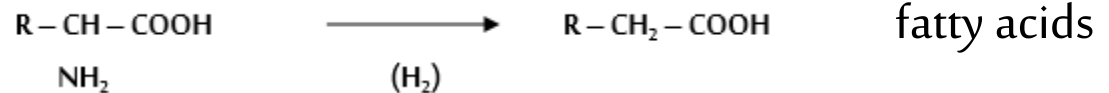


2- Deamination

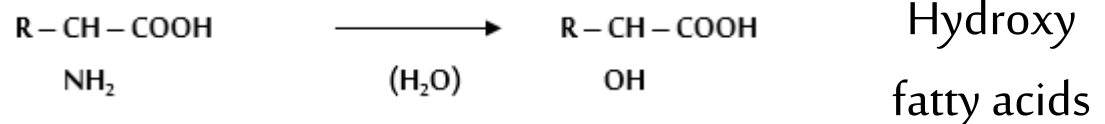
a. Oxidative deamination



b. Reductive deamination



c. Hydrolytic deamination

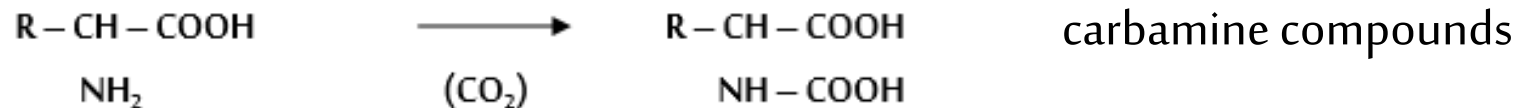


Properties of amino acids

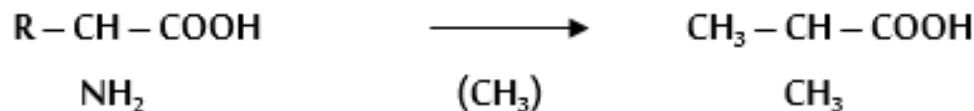
II. Chemical properties of amino acids:

A. Reactions due to presence of **amino group** of amino acids:

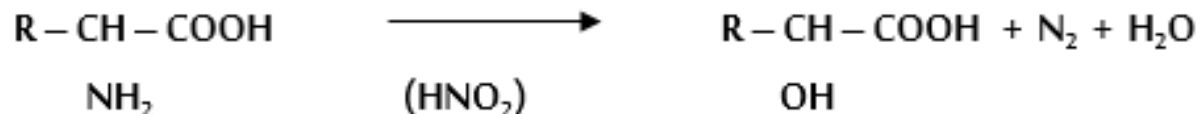
3- Carboxylation



4- Methylation



5-Action of nitrous acid

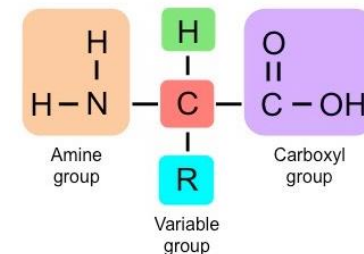


fatty acid

Properties of amino acids

II. Chemical properties of amino acids:

B. Reactions due to presence of carboxylic group of amino acids



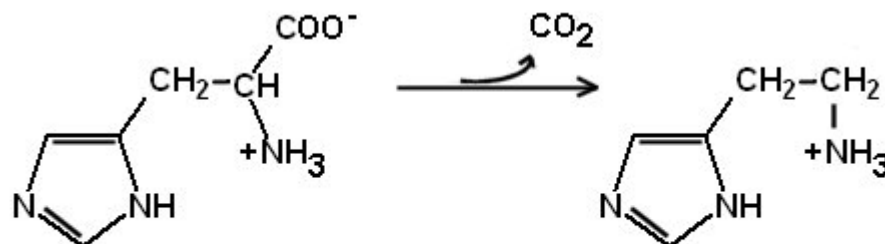
1- Salt formation



2- Decarboxylation



Amine



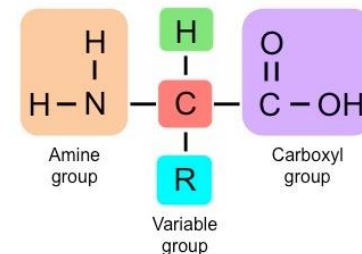
Histidine

Histamine

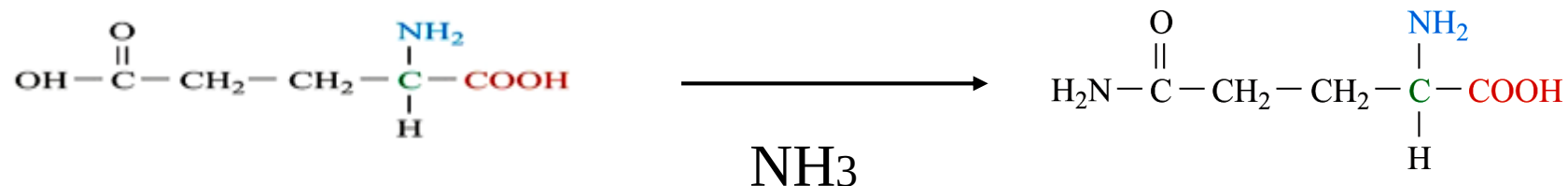
Properties of amino acids

II. Chemical properties of amino acids:

B. Reactions due to presence of carboxylic group of amino acids



3- Esterification

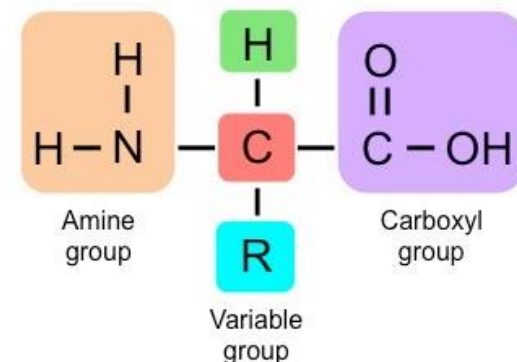


Properties of amino acids

II. Chemical properties of amino acids:

C. Reactions due to presence of **side chain** of amino acids

(Color reactions of proteins)



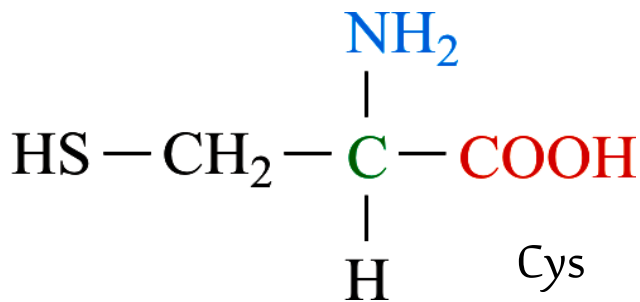
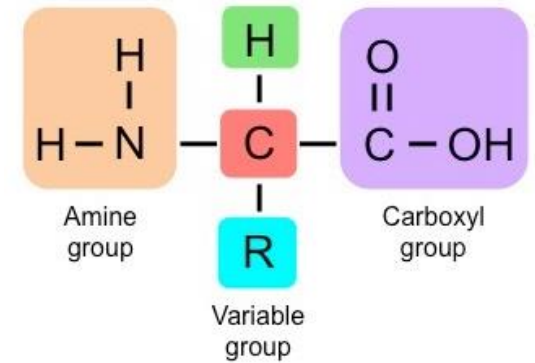
Reaction name	Principle	Resulting color
Rosenheim	indole group of tryptophan + conc. sulfuric acid + Rosenheim's reagent	Purple color
Millon	phenolic group of tyrosine + Millon's reagent	Red color after boiling.
Xanthoproteic	phenyl group of phenylalanine + conc. sulfuric acid	orange color
Sulfur	sulfur of cysteine or cysteine + lead acetate	Black color

Properties of amino acids

II. Chemical properties of amino acids:

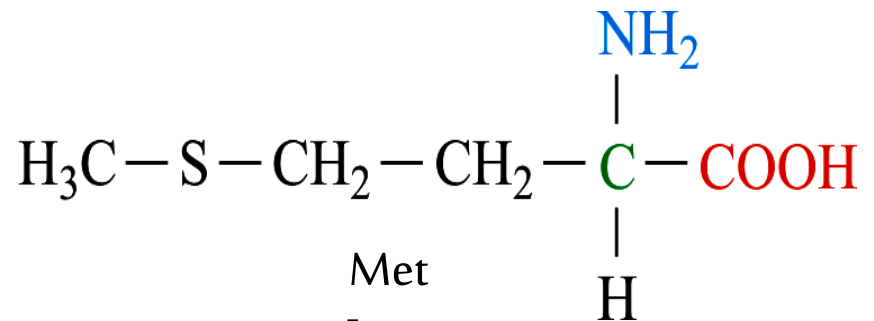
C. Reactions due to presence of **side chain** of amino acids

(Color reactions of proteins)



Black color

Sulfur test



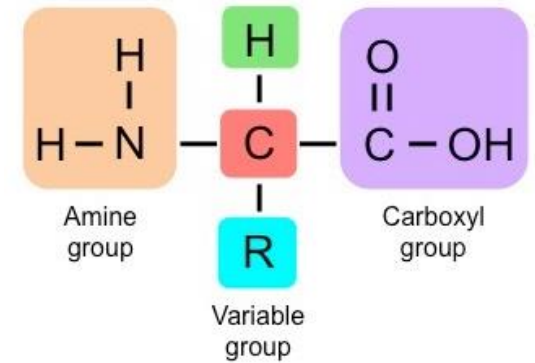
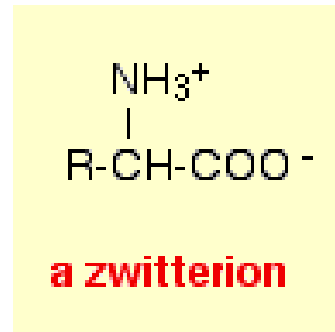
No Black color

Properties of amino acids

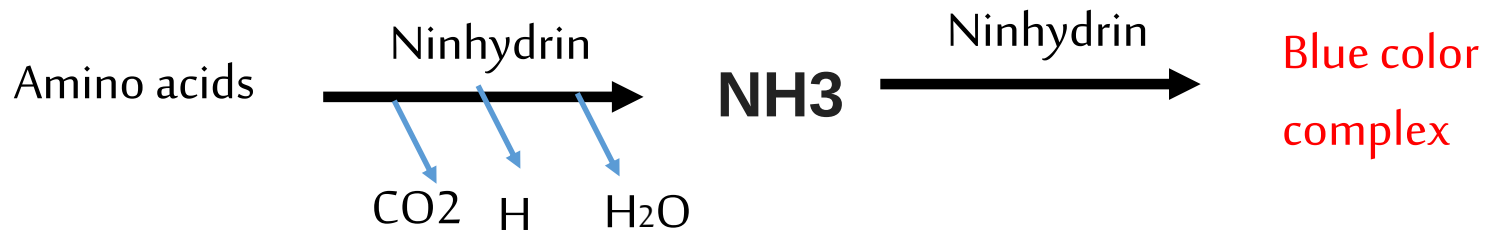
II. Chemical properties of amino acids:

C. Reactions due to presence of both COOH and NH₂ groups

1- Zwitterion formation



2- Reaction with ninhydrin



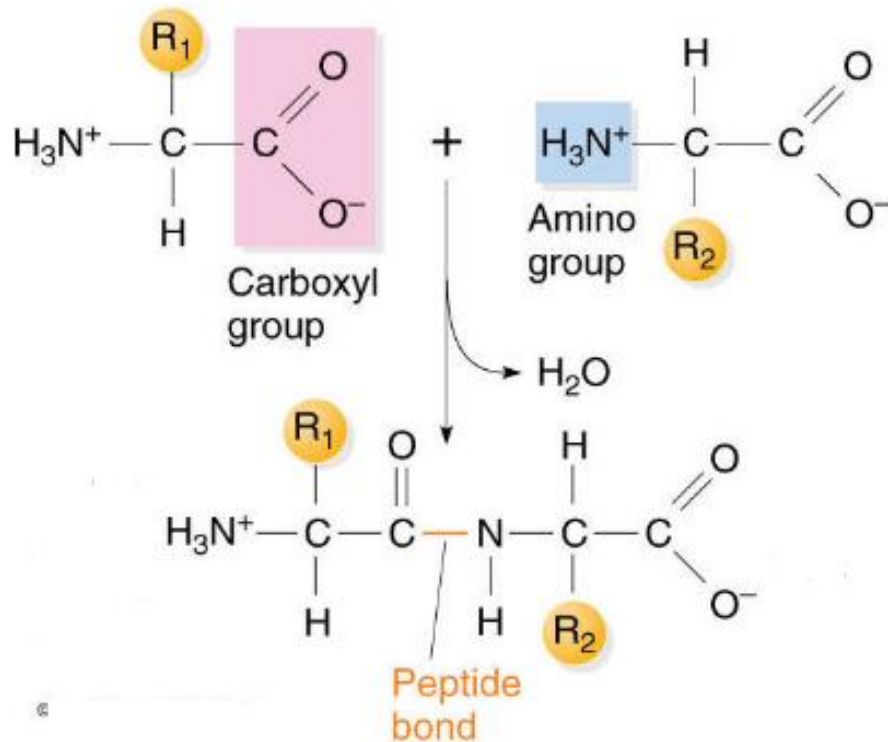
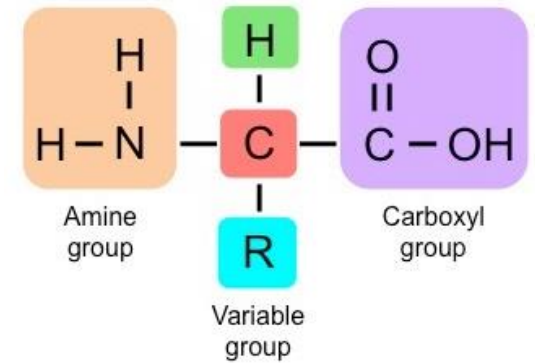
- The intensity of the blue color produced is used as a measure of the amount of amino acid present
- **Therefore**, Ninhydrin is used in **amino acid analysis of proteins**
- All amino acids, **except** proline, are hydrolyzed and react with ninhydrin

Properties of amino acids

II. Chemical properties of amino acids:

C. Reactions due to presence of both COOH and NH₂ groups

3 - Formation of **peptide** bonds

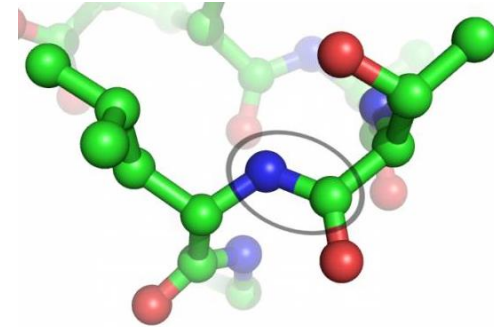


Peptide bond formation is
a **dehydration synthesis reaction**
(also known as a **condensation reaction**)

Peptides

A **peptide** is short chains of amino acids linked by peptide (amide) bonds

- If 2 amino acids are linked together they form a dipeptide,
- if 3 amino acids are together linked they form tripeptide
- Polypeptides may contain more than 10 and up to 100 amino acids



What Is the Difference Between a Peptide and a Protein?

- Peptides are **smaller** than proteins.

peptides are defined as molecules that consist of between **2 and 50** amino acids,
whereas proteins are made **up of 50 or more** amino acids

Peptides

Examples of physiologically active peptides

➤ Peptide hormones

i.e. insulin (51), glucagon (29), angiotensin I (10), angiotensin II (8) and gastrin (17)

➤ Antithrombotic peptides

- Inhibit the blood platelet aggregation
- i.e. Casoplatelins (bioactive peptides from bovine casein)

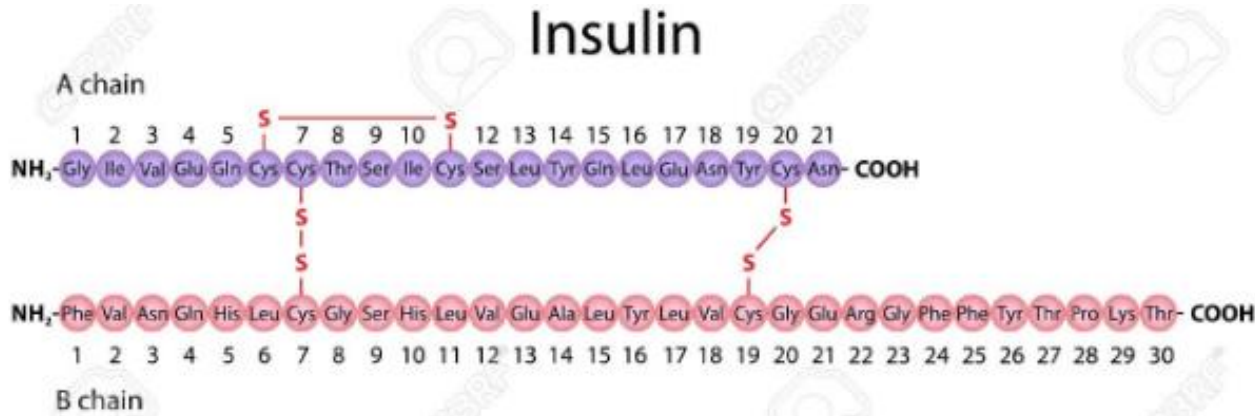
➤ Antioxidative peptides

- Inhibit lipid autooxidation process
- i.e. L-carnosine, Glutathione

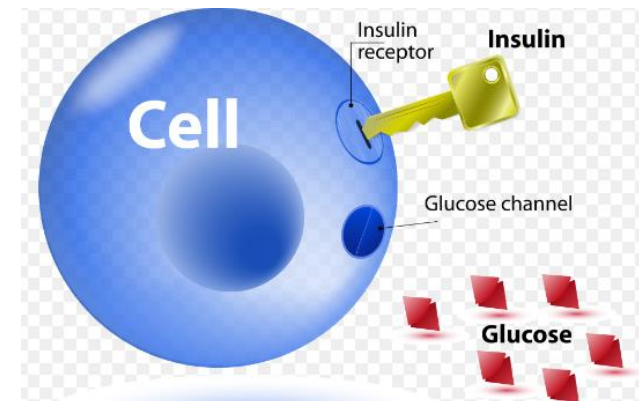
➤ Neuropeptides

- act as neurotransmitters
- i.e. Acetylcholine, and GABA

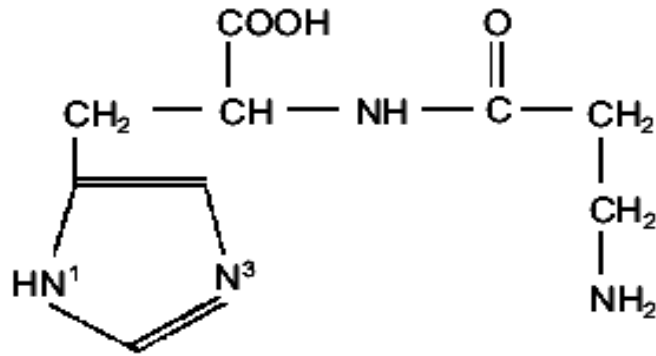
Insulin



- is a peptide hormone produced by **beta cells of the pancreatic islets**
- Function is to **maintain normal blood glucose level** (keeps your blood sugar level from getting too high (hyperglycemia) or too low (hypoglycemia))
 - Insulin is described as a “**key**,” which unlocks the cell to allow sugar to enter the cell and be used for energy.

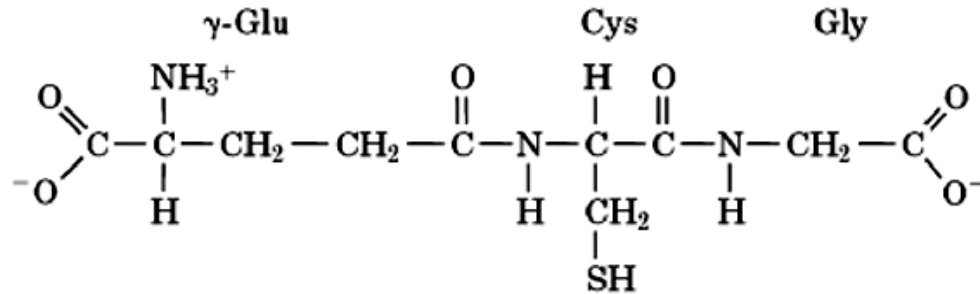


L-carnosine

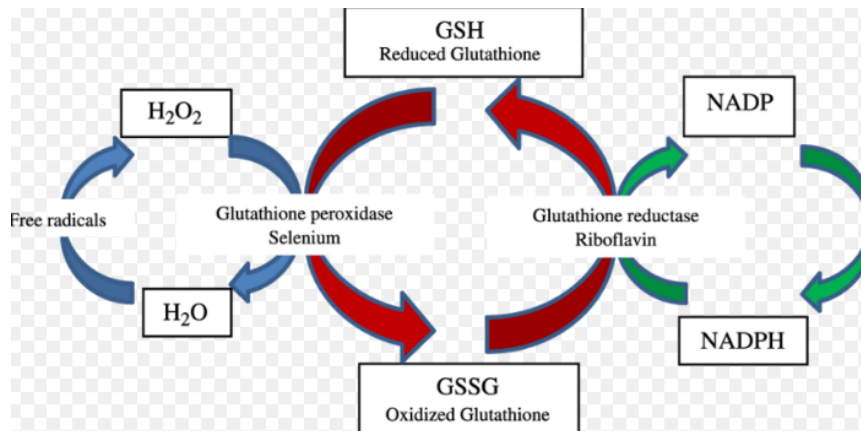


- Naturally occurring dipeptide made by chemical combination of β -alanine and histidine.
- Present in meat, fish and poultry.
- Can be synthesized by carnosine synthetase enzyme in the **brain and muscle**.
- **Biological importance:**
 1. Buffering function:
 2. Antioxidant function: inactivate reactive oxygen species and scavenge free radicals.
 3. Therapeutic uses: prevent signs of aging, improve muscle strength and exercise performance

Glutathione (GSH)



- a **tripeptide** comprised of three amino acids (cysteine, glutamic acid, and glycine)
- Produced naturally by the liver.
- Found in fruits, vegetables, and meats
- Can be present in reduced (GSH) and oxidized (GSSG) forms



Glutathione (GSH)

➤ Biological importance:

1. Powerful **antioxidant** (reducing) agent : preventing damage to important cellular components caused by reactive oxygen species
2. It activates many enzymes e.g. Glutathione peroxidase
3. Prevents hemolysis of the RBCs
4. It inactivate insulin hormone
5. Prevent rancidity of fat
6. Help in amino acids absorption

The background of the image is a vibrant cosmic scene featuring a nebula with swirling patterns of purple, blue, and yellow. Numerous small, bright stars are scattered throughout the dark space, adding depth and texture to the overall composition.

SEE YOU

NEXT WEEK!